Hemant Kumar

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PROFESSIONAL SUMMARY

Backend Engineer with nearly 2 years of experience building scalable Node.js backend systems using PostgreSQL, Redis, BullMQ, and MongoDB. Experienced in analytics platforms, asynchronous processing, database optimization, Docker, Kubernetes, and GitOps.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Java, SQL

Backend: Node.js, Express.js, Koa, REST APIs, WebSockets (Socket.IO)

Architecture: Microservices, Event-Driven Architecture, Multi-Tenancy, Caching

Databases & Messaging: PostgreSQL, MongoDB, Redis (Caching, Pub/Sub), BullMQ, RabbitMQ

DevOps & Cloud: Docker, Kubernetes, GitLab CI, ArgoCD, Oracle Cloud (OCI), Linux

Tools: Git/GitHub, Postman, Swagger, Jira

WORK EXPERIENCE

Software Developer

Enterprise Bot, Bengaluru, India *September 2024 - Present*

- Independently architected and built **Pulse from scratch**, a multi-tenant analytics and quota-management platform used by partners, clients, and internal management to monitor product usage, analyze team and environment performance, support usage-based billing and automate quota and usage notifications.
- Built a session-level pre-aggregation and batched backfill pipeline over datasets containing **400M+ raw events**, producing **7M+ consumption records** across **5+ years** of chat, voice, email, token, and LLM usage data.
- Reduced long-range PostgreSQL dashboard response time from approximately **14 seconds to under 1 second** through pre-aggregation, selective filtering, composite indexing, statistics tuning, and query-plan optimization.
- Architected an asynchronous alert-processing pipeline using **Node.js, BullMQ, Redis, and PostgreSQL**, with scheduled workers for quota and rule-based alert evaluation, retry handling, and notification processing.
- Built asynchronous streaming **CSV/ZIP export pipelines for 7M+ aggregated reporting records**, with progress tracking and object-storage delivery for large reporting workloads.
- Built and optimized **15+ PostgreSQL queries** powering assistant analytics metrics for conversation volume, engagement, and usage trends using CTEs, window functions, composite indexes, EXPLAIN ANALYZE, statistics tuning, and selective JIT configuration.
- Containerized backend services with Docker and worked with **Kubernetes, GitLab CI, and ArgoCD** to build, deploy, and promote application changes across environments.

PROJECTS

Chatty — Real-Time Messaging System *(Node.js, JavaScript, Socket.IO, Redis, MongoDB)* **GitHub**

- Architected and developed a real-time messaging backend supporting **1-on-1 conversations, presence, delivery status, and seen-status tracking**.
- Integrated the **Redis Socket.IO adapter** to support horizontal scaling and synchronize real-time events across multiple backend instances.
- Implemented offline synchronization and reconnect handling to ensure missed messages are delivered when users reconnect.

AlertOps — Queue-Based Alerting & Escalation System *(Java, Spring Boot, PostgreSQL, RabbitMQ, Redis, JPA)* **GitHub**

- Built a multi-tenant alerting platform with JWT-based access control, configurable alert flows, time-based escalations, and Redis-backed caching.
- Implemented an alert-flow engine using **optimistic locking** to handle concurrent workflow modifications without data loss.
- Engineered delayed escalation using **RabbitMQ delayed queues** and designed retry-safe processing with **at-least-once delivery and idempotent consumers**.

EDUCATION

Arya College of Engineering & IT

Jaipur, India

B.Tech in Computer Science & Engineering *Sep 2020 - May 2024*

CERTIFICATIONS

- Red Hat Certified System Administrator (RHCSA) — 284 / 300